

Graphing Proportional Relationships

Answer Key

Grades 6–7 Ratios and Proportional Relationships

Quick Answers

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|-------------|--------------|------------|--------|
| 1. 8, 8 | 4. 2.5 | 7. 3, 2 | 10. 18 |
| 2. 2.5, 2.5 | 5. 8 | 8. 16, 4 | |
| 3. 3 | 6. 5, 4, 3.5 | 9. 3.5, 42 | |
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Curriculum

Level: Grades 6–7 Ratios and Proportional Relationships

7.RP.A.2 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 20 tagged checks.

Common wrong answers

3. If they got 0.333333: read the point backwards and divided time by distance

1. Dog-walking pay: plot the table and find the unit rate.

Step 1. The three points are (0, 0), (3, 24) and (5, 40). Plotted, they lie on one straight line, and that line starts at the origin — the mark of a proportional relationship.

Step 2. Unit rate from the origin to (3, 24): $k = \frac{24 - 0}{3 - 0} = 8$.

Step 3. Check between two points that are not the origin: $k = \frac{40 - 24}{5 - 3} = \frac{16}{2} = 8$. The same rate everywhere is what makes the graph a straight line through (0, 0).

Manual review: the plotted points and the drawn line are the student's own work — check that all three points are placed correctly and the line is ruled through the origin. Dev earns 8 dollars per hour.

$k = 8$

2. Rice cost: plot the table and find the cost of one kilogram.

Step 1. Points (0, 0), (2, 5) and (6, 15). All three lie on one line through the origin, so cost is proportional to mass.

Step 2. From the origin to (2, 5): $k = \frac{5}{2} = 2.5$.

Step 3. Between the two non-origin points: $\frac{15 - 5}{6 - 2} = \frac{10}{4} = 2.5$ — the same constant.

Manual review: inspect the student's plotted points and ruled line. One kilogram of rice costs 2.5 dollars.

$k = 2.5$

3. Cyclist graph: unit rate in kilometres per hour.

Step 1. The line goes through the origin, so the unit rate is $y \div x$ at any point on it. Use the marked point (5, 15).

Step 2. $k = \frac{15}{5} = 3$ kilometres per hour.

Common wrong answer: 0.33, from dividing hours by kilometres. Always put the quantity named by the y -axis on top.

$$k = 3$$

4. Paint graph: write the equation.

Step 1. A line through the origin has the form $y = kx$, so the only thing to find is k .

Step 2. The marked point is (4, 10), so $k = \frac{10}{4} = 2.5$ square metres per litre.

Step 3. Substituting the constant gives the equation $y = 2.5x$.

$$y = 2.5x$$

5. Barrel: how long to collect 12 litres.

Step 1. The equation is $y = 1.5x$, and the question fixes the output at $y = 12$, so solve $1.5x = 12$ for x .

Step 2. $x = \frac{12}{1.5} = 8$ minutes.

Step 3. The graph agrees: reading across from 12 litres meets the line above 8 minutes.

$$x = 8$$

6. Is the relationship proportional?

Step 1. Divide y by x in every column. If the relationship is proportional, all the quotients are the same number.

Step 2. $5 \div 1 = 5$, $8 \div 2 = 4$, $14 \div 4 = 3.5$.

Step 3. The three quotients are different, so there is no single constant of proportionality. The graph settles it too: the line is straight but it meets the y -axis at 2, not at the origin, so it is not a proportional relationship.

$$5, 4, 3.5 : \text{not proportional}$$

7. Two runners: both unit rates and who is faster.

Step 1. Both lines pass through the origin, so each unit rate is $y \div x$ at that line's marked point.

Step 2. Runner A through (4, 12): $\frac{12}{4} = 3$ metres per second. Runner B through (5, 10): $\frac{10}{5} = 2$ metres per second.

Step 3. The steeper line has the larger unit rate, so runner A is faster — by 1 metre per second.

$$A = 3, B = 2$$

8. Water tank: difference between two readings, and the unit rate.

Step 1. Read the marked points: at 1 minute the tank holds 4 litres, at 5 minutes it holds 20 litres.

Step 2. Difference: $20 - 4 = 16$ litres more.

Step 3. Unit rate: $\frac{20 - 4}{5 - 1} = \frac{16}{4} = 4$ litres per minute, which matches $y \div x$ at either point.

$$16, k = 4$$

9. Bottling machine: unit rate, then a prediction off the graph.

Step 1. The marked point is $(4, 14)$ and the line goes through the origin, so $k = \frac{14}{4} = 3.5$ hundreds of bottles per hour.

Step 2. The equation is $y = 3.5x$.

Step 3. At $x = 12$ hours: $y = 3.5 \times 12 = 42$ hundreds of bottles. The drawn axis stops at 6 hours, which is exactly why the equation is worth having.

$k = 3.5, y = 42$

10. Printer: draw the graph, then find the time for 45 pages.

Step 1. 5 pages every 2 minutes is a unit rate of $\frac{5}{2} = 2.5$ pages per minute, so the equation is $y = 2.5x$ and the graph is the line through $(0, 0)$ and $(20, 50)$.

Step 2. Set the output to 45 pages: $2.5x = 45$.

Step 3. $x = \frac{45}{2.5} = 18$ minutes. On the graph, 45 pages sits between the gridlines at 40 and 50, above 18 minutes.

Manual review: the drawn line is the student's own work — it must start at the origin and pass through $(4, 10)$ and $(20, 50)$.

$x = 18$
